

Anirban Bhaduri

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Columbia, South Carolina, U.S.A.

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Education

University of South carolina

Doctorate in Philosophy (Ph.D.)(Advisor: Dr.Matthew Ballard)

January 2021 – Present

Columbia, South Carolina,U.S.A.

University of Calcutta

Masters in Science (M.Sc.)

August 2017 – June 2019

Kolkata, West Bengal, India

University of Calcutta

Bachelor of Science (B.Sc.)

August,2014 – July,2017

Kolkata, West Bengal, India

Research Interests

My Research interest lies in the intersection of Derived categories, Algebraic Geometry, Representation Theory, Non-Commutative and Homological Algebra.

Publications and Preprints

- Bhaduri A., Dey S., Lank P. *A note on generation and descent for derived categories of noncommutative schemes*—submitted
- Bhaduri A., Goldberg I., Robotis A.A. *Dimensions Theory of Noncommutative Curves*—submitted.
- Bhaduri A., Davidov Y., Faber E., Honigs K., McDonald P., Overton-Walker C.E., Spence D. *An Explicit Derived McKay Correspondance For Some Complex Reflection Group of Rank Two*—submitted.
- Ballard M.,Bhaduri A. *Orlov Spectra for Weighted Projective Stacks*—In Preperation.

Other Experience

Graduate Teaching Assistant

January,2021 – December,2021

University of South Carolina

Graduate Instructional Assistant

January,2022 – Present

University of South Carolina

Graduate Peer Mentor

January,2023 – April,2024

University of South Carolina

Presentations and Talks

Dimension Theory of Noncommutative Curves – Graz Algebra and Geometry Seminar, Fall 2024
Department of Mathematics, University of Graz

Dimension Theory of Noncommutative Curves – AGNES 2024, Fall 2024
Department of Mathematics, Dartmouth College

On the derived category coherent sheaves of Projective line – Graduate Colloquium, Spring 2023
Department of Mathematics, University of South Carolina

An introductory talk on Quivers and Representation Theory – AGCA Seminar, Spring 2022
Department of Mathematics, University of South Carolina

Teaching

Instructor

University of South Carolina

Columbia, South Carolina, U.S.A.

- MATH 152 — Calculus Workshop Fall 2024
- MATH 170 — Finite Mathematics Spring 2024
- MATH 115 — Pre-Calculus Mathematics Fall 2023
- MATH 142 — Calculus II Summer 2023
- MATH 122 — Calculus for Business Administration Spring 2023
- MATH 111 — Basic College Mathematics Fall 2022
- MATH 141 — Calculus I Summer 2022
- MATH 115 — Pre-Calculus Mathematics Spring 2022

Teaching Assistant

University of South Carolina

Columbia, South Carolina, U.S.A.

- MATH 141 — Calculus I Fall 2021
- MATH 141 — Calculus I summer 2021
- MATH 141 — Calculus I Spring 2021

Conferences and Workshops

Introductory Workshop: Noncommutative Algebraic Geometry February 5-9, 2024. Simons Laufer Mathematical Sciences Institute, formerly Mathematical Sciences Research Institute.

Connections Workshop: Noncommutative Algebraic Geometry February 1-2, 2024. Simons Laufer Mathematical Sciences Institute, formerly Mathematical Sciences Research Institute.

Mathematics Research Communities - 2023 June 4 -10, 2023. American Mathematical Society.

Concluding Conference of Simons Collaboration on Homological Mirror Symmetry April 25-28, 2023. Simons Center for Geometry and Physics, Stony Brook University.

Derived Categories, Moduli Spaces and Hyperkahler Varieties Minischool(July 30-31,2022) and Conference(August 1-5,2022). University of Michigan.

Homological Mirror Symmetry April 4-8, 2022. University of Miami, 5030 Brunson Drive, Cosford Cinema.

Analysis, Algebra, Number Theory and Probability Workshop February, 2019. Department of Mathematics, Rahara Ramkrishna Mission Vivekananda Centenary College, Kolkata, West Bengal, India.

Organization

Derived And (Non)Commutative Enthusiasts (D.A.N.C.E.)- Spring 2025, Online Seminar
Algebraic Geometry Commutative Algebra and Number Theory Graduate Student Seminar- Fall 2023, University of South Carolina

Mathematical Research Communities Sectional at Joint Mathematics Meeting- January 2024